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ENBC 332

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Homework 8 Discussion and Outputs

Problem 2:

Linearity, Normality of Residuals, and Constant Variance of Errors were all satisfied based on the data generated. The “Residuals vs Fitted” plot consists of points scattered randomly and has no pattern, shown by the red line that stays approximately 0 which proves the relationship between BMI and SBP as linear. Normality of the Residuals can be proven using the “Normal Q-Q” plot where all points fall along the dashed diagonal line, signifying a normal distribution indicating that the residuals are normally distributed. Constant Variance can be proven through the “Scale-Location” plot. The points are randomly scattered, and the vertical spread remains consistent across all fitted values which indicate that the variance of the residuals is constant.

```

> # Data
> ad_num <- c(1, 2, 3, 4, 5, 6)
> mon_sal <- c(3, 5, 7, 10, 12, 14)
> # Part A
> df <- data.frame(ad_num, mon_sal)
> plot(df$ad_num, df$mon_sal, main="Scatter Plot", xlab="Number of Advertisements", ylab="Monthly Sales in $1000")
> # Part B
> lin_reg <- function(x_data, y_data) {
+   x_mean <- mean(x_data)
+   y_mean <- mean(y_data)
+
+   x_diff <- x_data - x_mean
+   y_diff <- y_data - y_mean
+
+   num <- sum(x_diff * y_diff)
+   den <- sum(x_diff^2)
+
+   m <- num / den
+
+   return(c(m, y_mean - m*x_mean))
+ }
> lin_reg_custom <- lin_reg(df$ad_num, df$mon_sal)
> lin_reg_func <- lm(mon_sal ~ ad_num, data=df)
> slope <- lin_reg_custom[1]
> y_intercept <- lin_reg_custom[2]
> cat("Custom Function Values\nSlope:", slope, "\nY-Intercept:", y_intercept, "\n")
Custom Function Values
Slope: 2.257143
Y-Intercept: 0.6
> summary(lin_reg_func)

Call:
lm(formula = mon_sal ~ ad_num, data = df)

Residuals:
    1     2     3     4     5     6 
0.1429 -0.1143 -0.3714  0.3714  0.1143 -0.1429 

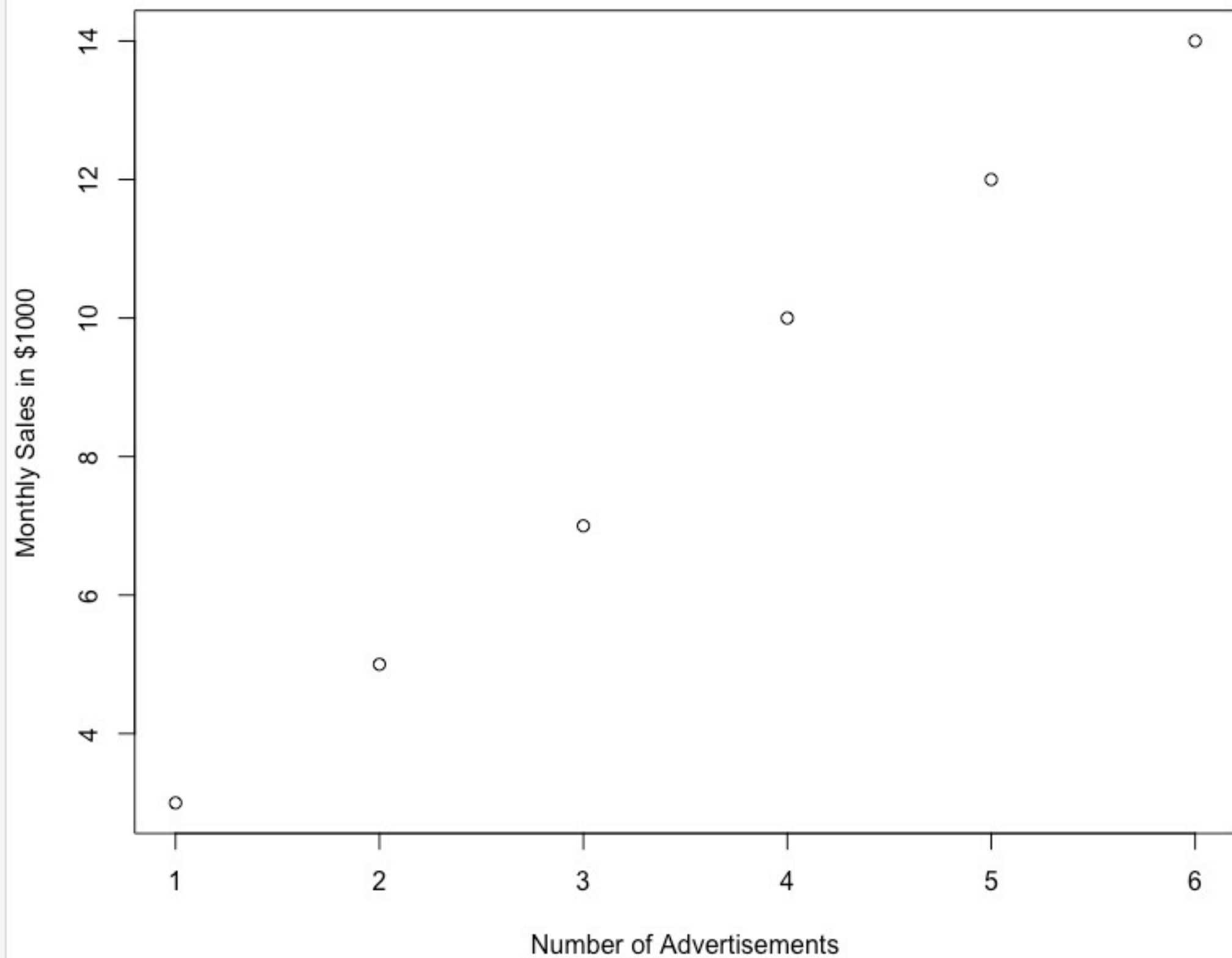
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.60000    0.27255   2.201   0.0925 .
ad_num       2.25714    0.06999  32.252 5.51e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.2928 on 4 degrees of freedom
Multiple R-squared:  0.9962,    Adjusted R-squared:  0.9952 
F-statistic: 1040 on 1 and 4 DF,  p-value: 5.51e-06

> # Part C
> cat("Sale Prediction for 4 Advertisements in a Month:", slope * 4 + y_intercept,
+     "\nSale Prediction for 6 Advertisements in a Month:", slope * 6 + y_intercept)
Sale Prediction for 4 Advertisements in a Month: 9.628571
Sale Prediction for 6 Advertisements in a Month: 14.14286
> # Part D
> predicted_values <- slope * df$ad_num + y_intercept
> pre_val_mean <- mean(predicted_values)
> SST <- sum((df$mon_sal - mean(df$mon_sal))^2)
> SSR <- sum((predicted_values - mean(df$mon_sal))^2)
> SSE <- sum((df$mon_sal - predicted_values)^2)
> R2 <- SSR / SST
> cat("R Squared Value:", R2)
R Squared Value: 0.9961692

```

Scatter Plot



```

> # Data
> set.seed(123)
> BMI <- rnorm(300, mean = 25, sd = 5)
> SBP <- 90 + 2.5 * BMI + rnorm(300, mean = 0, sd = 8)
> # Part A
> df <- data.frame(BMI, SBP)
> lin_reg <- lm(SBP ~ BMI, data=df)
> summary(lin_reg)

```

Call:
 lm(formula = SBP ~ BMI, data = df)

Residuals:

Min	1Q	Median	3Q	Max
-21.9754	-4.9021	0.4433	5.1154	20.5909

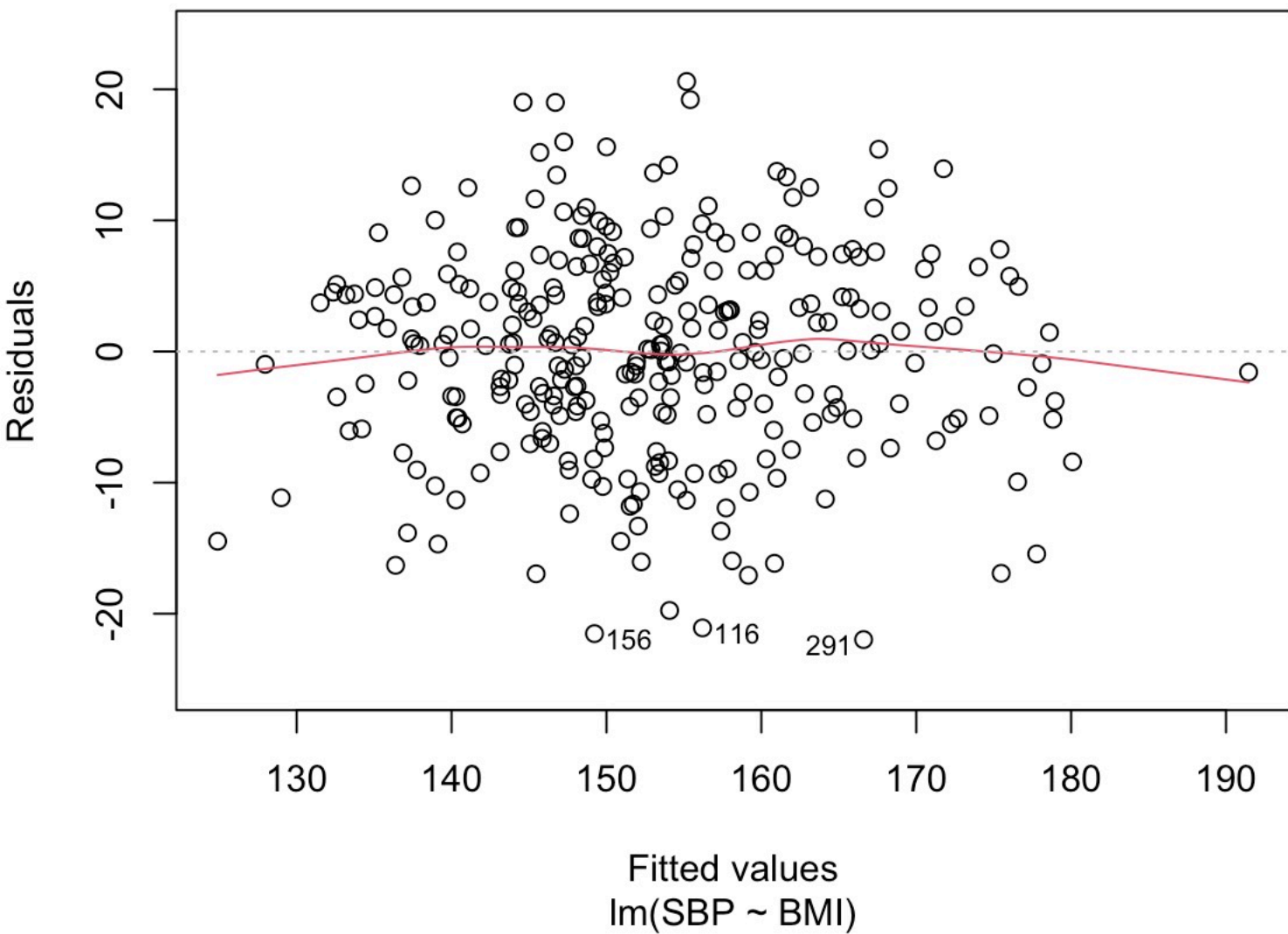
Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	92.62855	2.47733	37.39	<2e-16 ***
BMI	2.39847	0.09673	24.80	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

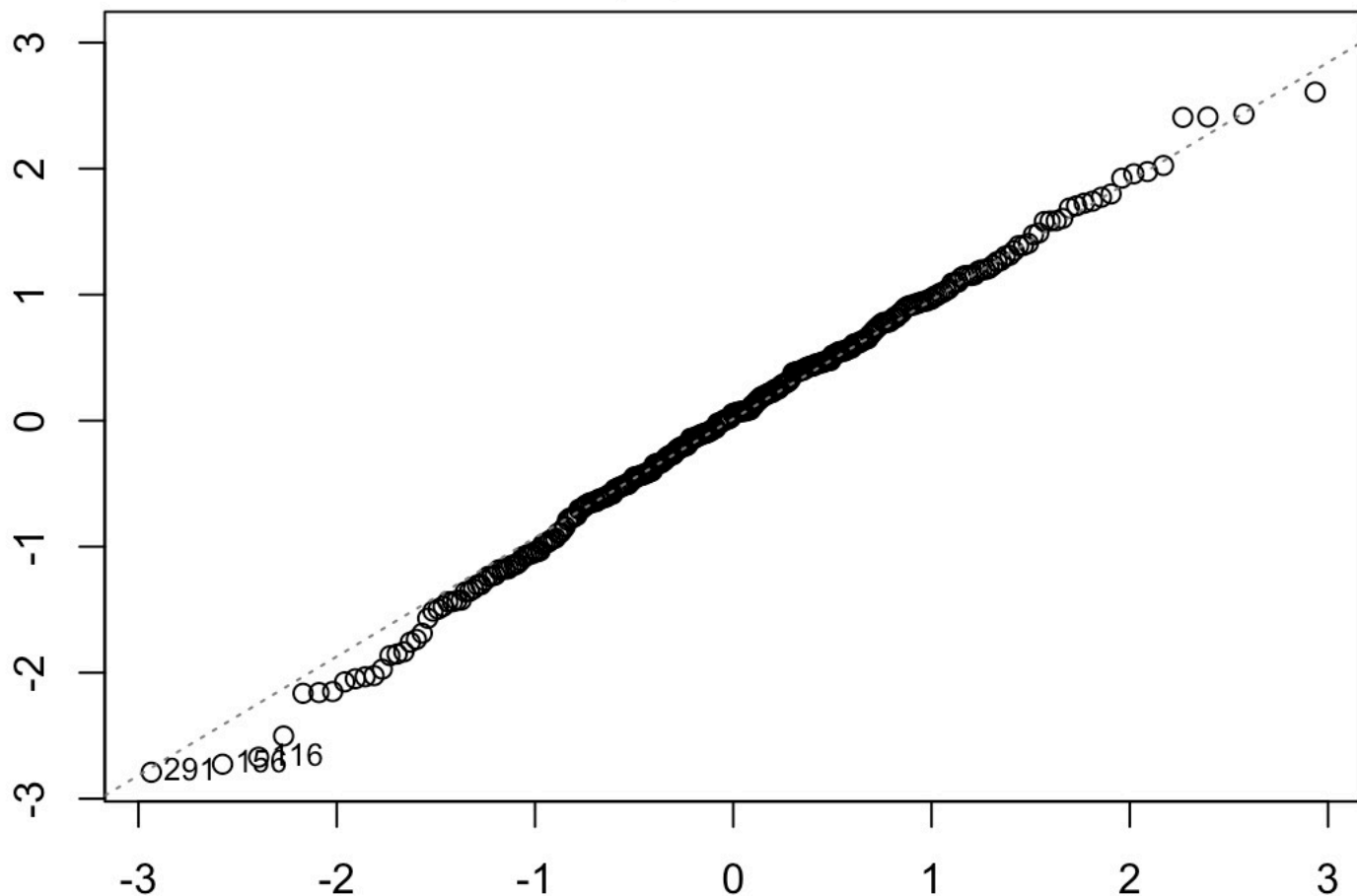
Residual standard error: 7.91 on 298 degrees of freedom
 Multiple R-squared: 0.6735, Adjusted R-squared: 0.6724
 F-statistic: 614.8 on 1 and 298 DF, p-value: < 2.2e-16

Residuals vs Fitted



Q-Q Residuals

Standardized residuals



Theoretical Quantiles
 $\text{lm}(\text{SBP} \sim \text{BMI})$

